2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSE-1103

(VLSI Design)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer *question no.* 1 and *question no.* 2 and *any four* questions from the rest.

1. Answer *any five* questions : 2×5

(a) What is feature size? How is it associated with device count?

(b) Mention two usual factors and two high performance issues involving deep-submicron design.

(c) Explain the transfer characteristic of CMOS inverter circuit.

(d) Why is it necessary to reduce the chip area?

(e) Suggest the practices usually applied to solve if most of the problems in VLSI physical design are computationally hard.

(f) Draw the 2-to-1 multiplexer using transmission gates, and provide its function table.

(g) Explain whether a 3-input CMOS NAND gate and a 3-input CMOS NOR gate perform identically.

2. Answer *any five* questions : 4×5

(a) State *Shannon's Expansion Theorems,* and show that the size of a switch matrix for realizing any 3-variable function is at most 4x4 using only nMOS combinational network. Give an example using stick diagram.

(b) Draw a CMOS 4-input AND gate circuit and present its function table.

(c) Clearly describe how a transmission gate functions when the control input is high.

(d) Show that at most two transmission gates suffice to acquire any 2-variable function in the form of CMOS combinational network. Give at least two examples.

(e) In terms of limitations and novelties, relatively compare pMOS, nMOS and CMOS circuits.

(f) Relatively compare custom design and standard cell design among the different design styles available.

(g) Justify if the area minimization problem and the wire length minimization problem in channel routing are distinct and different problems.

3. (a) State the concept behind *Group Migration Algorithms* and show how *Kernighan-Lin algorithm* works for the following graph with initial partitions $A = \{p, q, r, s\}$ and $B = \{w, x, y, z\}$: Vertices p, q, w, x form a complete graph, vertices r, s, y, z form a complete graph and vertices q and y, and r and x are connected by edges.

(b) Briefly distinguish between simulated annealing and simulated evolution techniques in devising algorithms for most of the problems in VLSI physical design. (1+5)+4

4. Define partitioning problem in VLSI physical design and state its significance. Briefly describe the formulation of partitioning problem such that the associated constraints and objective functions could be taken care of while designing any partitioning algorithm. 2+8

5. (a) Mention necessary assumptions, if any and compute a lower bound on the number of tracks of the following channel instance :

0 5 6 5 6 0 2 3 0 3 0 4

5 2 0 5 2 6 3 1 4 4 1 0

(b) Define doglegging. Show how doglegging helps in reducing channel area for the following channel instance and compute the percentage of reduction in channel area. 4+(1+5)

3 3 0 4 2 2 0

0 4 4 2 0 1 1

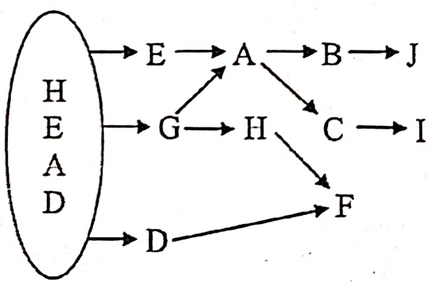
6. (a) Realize the following circuit that comprises one pMOS transistor Q_1, one nMOS transistor Q_2, one 2-input CMOS NAND gate, one 2-input CMOS NOR gate and one CMOS inverter. The inputs to the circuit are X and Y such that the gate input to Q_1 is $f_p = (X' \; Y)'$ and that to Q_2 is $f_n = (X+Y)'$. Moreover, the source of Q_1 is connected to V_{DD} whereas that of Q_2 is grounded, and the drains of the transistors are tied together from where the output, F is obtained.

(b) Draw stick diagrams of (i) an nMOS inverter, (ii) a CMOS inverter and (iii) a 3-input CMOS NAND gate. 6+(1+1+2)

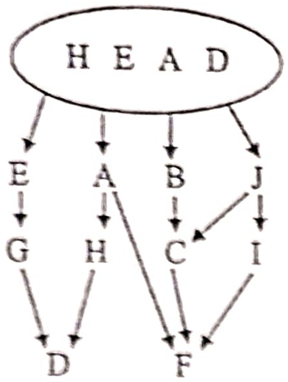
7. (a) Mention how the factor(s) to be taken care of while the formulation of placement problem in devising usual and high performance circuits. How *Steiner tree* and *minimum spanning tree* play their role in designing placement algorithms? What do you mean by *force directed placement* ?

(b) Consider the following two diagrams based on the relative positions of rectangular blocks, one from left to right and the other from top to bottom. Assign all the blocks, as have been directed in the diagrams, without any wastage, on a rectangular chip floor. (3+2+1)+4

From left to right :



From top to bottom :



8. Define the term floorplanning in VLSI physical design. Define and exemplify different types of floorplanning. What do you mean by *hierarchical floorplan of order k* ? Give an example of such a floorplan where the value of *k* is at least 7. Define *floorplan tree* and draw the corresponding floorplan tree. 1+3+1+2+3